

Human-like autonomy emerges from guided self-play

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Abstract

DC: rewrite We enable autonomous agents to navigate in a robust and human-like manner by mixing the capabilities and skills learned through self-play with behaviors derived from human data. Self-play agents collide rarely among themselves but frequently violate road norms. Learning from human data has the opposite problem: it creates natural-looking behavior but has collision rates well above human experts. To acquire the best of both settings, we train agents through self-play to flexibly follow expert guidance, allowing them to take suggestions from pre-trained imitation models while ignoring advice that might lead to a colliding state. We evaluate our approach on the Waymo Open Sim Agent Challenge, where it achieves competitive performance while offering improved behavioral diversity, controllability, and robustness. Videos, code, and models are available at <https://daphne-cornelisse.github.io/guided-self-play>.

1 Introduction

Safely deploying intelligent robots in the real world requires systems capable of effectively modeling and interacting with humans in the wild. Autonomous driving offers a concrete example. The safety and social acceptance of a deployed autonomous vehicle (AV) hinges on understanding how people behave in relatively unstructured, naturalistic settings. Whether merging into fast-moving traffic or negotiating a four-way stop, AVs must operate within systems designed by and for humans. To succeed, they must not only *predict* and *plan* realistic interactions with human road users but also engage seamlessly with them.

A stepping stone on the road to human-compatible autonomous agents may be to develop *human-like autonomy in multi-agent simulation*¹ first. If we can achieve this, that would enable a safe, cost-effective, and scalable complement to real-world testing and data collection. In turn, it can accelerate the development of autonomous systems that interact effectively with humans.

One useful way to decompose the goal above is to view it along two axes: human-likeness and controllability.

Human-likeness, sometimes also referred to as realism, measures how closely an agent’s behavior resembles that of human road users. At a macroscopic level, we can ask whether the agent produces realistic distributions over key behavioral metrics, such as speed, acceleration, and distances to other road users. While no model captures human behavior perfectly (Nester, 1996), statistical alignment with real-world data can be sufficient to support meaningful evaluation and training (Montali et al., 2023, 2024).

Controllability refers to the ability of an engineer or researcher to systematically steer an agent’s behavior toward desired outcomes. This is typically achieved by defining a reward function that defines a hierarchy of costs or utilities. For example, to stress-test autonomous systems under diverse conditions, one might create a range of agent behaviors by adjusting parameters such as the penalty for collisions or the preference for maintaining a certain speed.

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¹Specifically, we refer to closed-loop simulation in the control theory sense: an agent’s actions influence the environment’s state, which in turn provides feedback affecting subsequent actions, including those of other agents.

The dominant approach to learning human-like behavior relies on imitation learning on large-scale datasets of driving demonstrations (Ngiam et al., 2021; Zhou et al., 2022, 2023; Zhang et al., 2023a). While imitation can yield natural-looking behavior, it is often brittle in multi-agent closed-loop settings that involve feedback, since compounding errors and out-of-distribution scenarios are likely to occur. Moreover, collecting and labeling enough high-quality demonstrations to cover long-tail behaviors is both difficult and expensive (Makansi et al., 2021; Sun et al., 2024).

In contrast, self-play reinforcement learning (RL) allows agents to learn through *simulated* interactions with evolving copies of themselves. This paradigm has long been successful in multi-agent settings such as strategic games (Silver et al., 2016; Jaderberg et al., 2019; Heaven, 2019; Vinyals et al., 2019), and has recently shown promise in autonomous driving as well (Cusumano-Towner et al., 2025; Cornelisse et al., 2025). Cusumano-Towner et al. (2025) demonstrate that, after training in simulation—penalizing collisions over the equivalent of [X] million miles (X days of simulated time), such agents can reach crash rates as low as once every 17.5 years.

However, a well-known limitation of self-play is that agents can converge to equilibria that, while effective, are incompatible with human behavior [CITE]. On the road, many different conventions can support effective coordination, but only a subset align with how people navigate traffic. This is reflected in the realism gap observed in self-play-trained driving agents: they achieve a relatively poor WOSAC realism score of 0.62 (Cusumano-Towner et al., 2025). Qualitatively, agents often weave through tight gaps or execute unnaturally aggressive merges — high-risk maneuvers that most human drivers would avoid.

To fully leverage the strengths of self-play in learning autonomous driving agents, a remaining piece of the puzzle is therefore understanding how to *guide the learning process* toward human-compatible equilibria. In this paper, we contribute a step in that direction. Rather than relying solely on imitation or pure self-play, our approach combines both: agents learn through closed-loop interaction, with guidance from human demonstrations.

We evaluate our approach on the Waymo Open Sim Agent Challenge (WOSAC) (Montali et al., 2024). Key results of our work are:

1. **Data efficiency:** Agents achieve state-of-the-art realism on the WOSAC benchmark (meta-score 0.76) using only 2% of the Waymo Open Motion training dataset (10k of 487k scenarios).
2. **Realism across the spectrum of guidance densities:** Realism remains consistent whether agents receive full trajectories or only endpoint guidance.
3. **Generalization:** Agents retain realism even when guidance is out-of-distribution

Collectively, our findings show that guided self-play is a viable path to learning realistic, highly controllable agents. This approach considerably reduces the need for real world human data collection, and it does not require large, advanced architectures.

2 Related work

Imitation learning Imitation Learning (IL) has emerged as a significant data-driven approach in the development of autonomous vehicles (AVs). At its core, IL involves training an agent to mimic the demonstrated expert actions within a given scene, by leveraging large-scale datasets (Caesar et al., 2019; Ettinger et al., 2021; Wilson et al., 2023) comprising environment observations paired with the corresponding real-world traffic behaviors. Such paradigm has been widely adopted in the trajectory prediction tasks, where the model learns to forecast the marginal future trajectories of agents in the scene (Salzmann et al., 2020; Zhao et al., 2021; Gu et al., 2021; Shi et al., 2022; Varadarajan et al., 2022; Nayakanti et al., 2023), or the joint scenario rollouts of multiple interactive agents (Ngiam et al., 2021; Zhou et al., 2022, 2023), based on their past trajectories and the surrounding environment. Recently, research community has been focused on developing realistic agent policies, which are capable of generating safe, consistent, and human-like behaviors in closed-loop for complex multi-agent scenarios (Chen et al., 2024). These agents have potential of serving the ego-vehicle’s planning modules, which generate trajectories from intermediate environment representations in a mid-to-end fashion (Bansal et al., 2018; Jiang et al., 2023a), or directly from the raw sensor data in an end-to-end manner (Hu et al., 2023; Weng et al., 2024). Beyond planning future actions for the ego-vehicle, these agents can also be used to

simulate the behavior of surrounding agents, enabling realistic and reactive simulations for AVs’ learning and evaluation (Montali et al., 2023). One prominent approach to learn these policies effectively is to use generative models that discretize the ground truth trajectories into tokens and train autoregressive models to predict the next token given the previous ones (Seff et al., 2023; Pillion et al., 2023; Wu et al., 2024). Another stream of methods utilize diffusion models (Ho et al., 2020), to learn the distribution of agent trajectories and environment features (Zhong et al., 2023b); Jiang et al. (2023b) or directly model the joint distributions of multiple agents in the scene (Zhong et al., 2023a; Huang et al., 2024; Jiang et al., 2024). These models have shown promise in generating diverse, realistic and controllable traffic scenarios through guidance, but they often require extensive computational resources and may not be suitable for real-time applications.

RL for self-driving and finetuning Self-play based approaches for learning agents; Driving / traffic (Cusumano-Towner et al., 2025; Cornelisse et al., 2025; Cornelisse and Vinitzky, 2024)

Finetuning (Peng et al., 2024; Tian and Goel, 2025)

Imitation-guided RL

3 Method

3.1 Dataset and simulator

We conduct our experiments in GPUDrive (Kazemkhani et al., 2024). GPUDrive is a data-driven, multi-agent, GPU-accelerated simulator that is compatible with naturalistic traffic scenarios. For this paper, we use the Waymo Open Motion Dataset (Ettinger et al., 2021). We use 10,000 randomly sampled scenarios from the train dataset to train the agents. We evaluate on two test sets: 1) 1,000 held-out scenarios from the WOMD validation dataset; 2) 229 hand-picked scenes from the interactive validation dataset that were specifically selected by Zhang et al. (2023b) based on their difficulty (interactiveness) and perfect ground-truth (there are no collisions due to labeling errors in the logs).

3.2 State and observation space

Every agent receives a lightweight but structured partial view of the scene through a local coordinate frame. Figure 1 illustrates the different types of information in the agent’s observation. We briefly discuss the modalities below and refer to Appendix B for more details. All features are normalized to be in the $[-1, 1]$ range based on predefined physical ranges.

Ego state Each agent receives basic information about its own state, including speed, size of the bounding boxes, and steering angle. Additional task-relevant features include the progress along the suggested route (the fraction of visible x, y positions that the agent has hit), the normalized remaining time in the episode, and the previous action taken. The latter is included to make the agent aware of its dynamics.

Partner observations The partner observation includes information about nearby agents ($\square$ ’s in Figure 1): Their relative position, velocity, orientation, and size.

Road map features The road graph observations comprise a sparse subset of the local road geometry within a fixed radius of $r = 50$ meters. These points are projected into the ego frame and downsampled using a polyline reduction algorithm to reduce redundancy along straight segments. Each point includes spatial and geometric attributes such as width, orientation, and type (e.g., lane centerline, road edge). We set the number of maximum observable road points to 186, as we empirically find that this is sufficient to perform well.

Guidance The guidance modality contains expert suggestions in the form of positions (marked as $\bullet$ ’s in Figure 1), reference headings, and speeds. We provide suggestions for the full scene, but transform them to always be

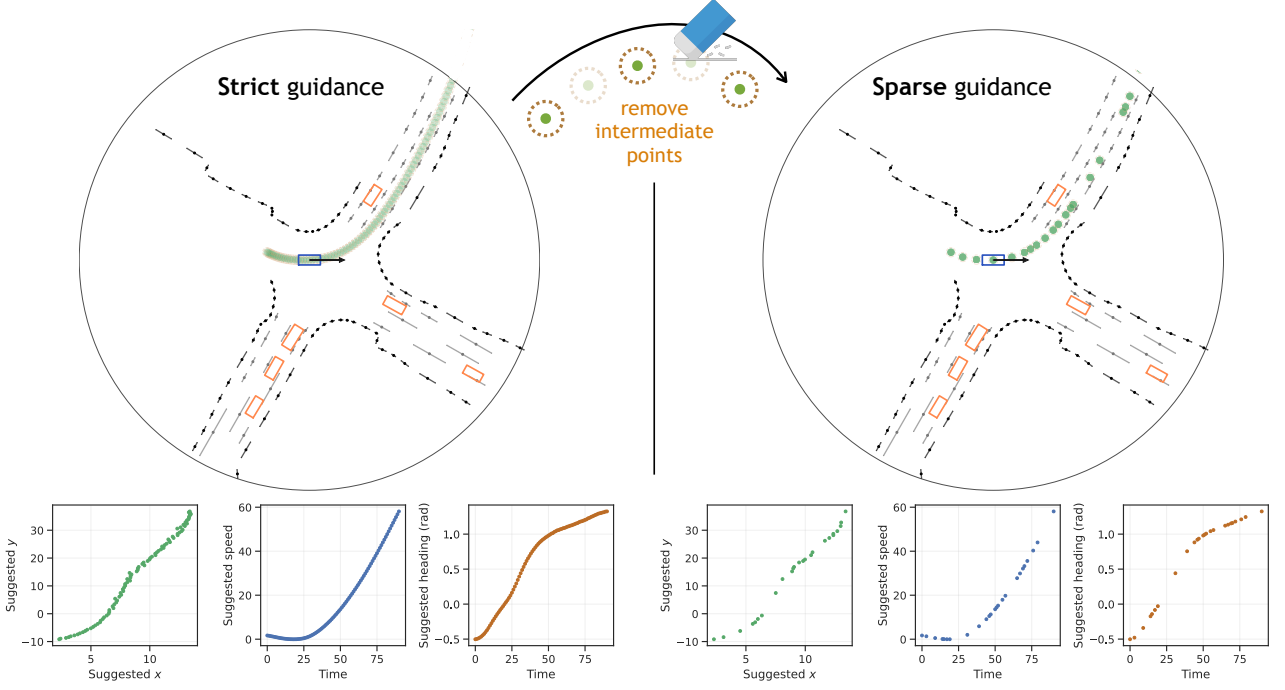


Figure 1: Overview of agent observation with guidance points. Agents receive a sparse representation of the road graph and other agents surrounding it within 50 meters. Agents also receive basic information about themselves, such as the length and width or their bounding box, and their current speed. Additionally, agents receive guidance points. These consist of $(x, y, s, \psi)_t$ tuples representing the suggested position, speed, and heading for the agent. At each step, the suggested guidance points are updated so that they are provided with respect to the current position of the agent. The number of guidance points per agent in the log replays varies. Each agent has at least one valid point and at most 91 (since the length of the discretized log playback trajectories in WOMD is 91). During training, we drop out a fraction of the guidance points. *Left*: We refer to the mode where an agent has a suggested point at every time step as **strict guidance**; *Right*: By dropping out a fraction of the points, we control the guidance density. This agent is an example of the observation at a dropout rate of 0.5. The lower plots display the time series of guidance positions, speed, and heading, whereas the top figure displays the guidance positions together with the rest of the information in the agent’s observation.

relative to where the agent is at a particular timestep.

$$\mathbf{o}_{gd}^i = \begin{bmatrix} x_1 & y_1 & s_1 & \psi_1 \\ x_2 & y_2 & s_2 & \psi_2 \\ x_3 & y_3 & s_3 & \psi_3 \\ \vdots & \vdots & \vdots & \vdots \\ x_T & y_T & s_T & \psi_T \end{bmatrix} \quad (1)$$

We drop out a large fraction of the suggestions during training (details in Section 3.5), where the last valid point is kept to ensure agents always have a point of reference. Here is an example of a guidance observation with dropout (we set dropped out values to -1.0):

$$\mathbf{o}_{gd}^i = \begin{bmatrix} x_1 & y_1 & s_1 & \psi_1 \\ \text{ } & \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } & \text{ } \\ x_4 & y_4 & s_4 & \psi_4 \\ \text{ } & \text{ } & \text{ } & \text{ } \\ \text{ } & \text{ } & \text{ } & \text{ } \\ \vdots & \vdots & \vdots & \vdots \\ x_n & y_n & s_n & \psi_n \end{bmatrix} \quad (2)$$

We smooth the speed and heading targets to reduce noise in the logs (Example of raw and smoothened traces in Figure 8). The guidance mode can be configured to either use offline logs replay from the Waymo Open Dataset or online predicted trajectories from the pre-trained diffusion model, Versatile Behavior Diffusion (VBD) Huang et al. (2024).

The full observation is provided as a flattened vector containing the ego state, partner features, road map points, and guidance signals:

$$\mathbf{o}_t^i = [\mathbf{o}_{\text{ego}}, \mathbf{o}_{\text{partner}}, \mathbf{o}_{\text{road}}, \mathbf{o}_{\text{gd}}]$$

3.3 Dynamics model and action space

Each agent i selects an action vector $\mathbf{a}_t^i = (\tilde{a}, \tilde{s}, \tilde{h})$, where $\tilde{a}$ and $\tilde{s}$ denote discrete acceleration and steering actions, respectively. The heading $\tilde{h}$ is fixed to zero for simplicity, as we assume agents have a full 360° field of view within a radius r —a design choice reflecting either privileged human perception or the AV sensors. Acceleration is discretized into 7 values within $[-4, 4]$, and steering into 13 values within $[-\pi/2, \pi/2]$, aligned with semi-realistic kinematic constraints. While the discrete action space remains the same across agent types, its interpretation varies: the same discrete index maps to different physical values depending on whether the agent is a vehicle, cyclist, or pedestrian. For instance, vehicles have acceleration bounds of $[-4, 4]$ m/s² and steering bounds of $[-1.57, 1.57]$ radians, whereas pedestrians operate within $[-1.5, 1.5]$ m/s² and $[-3.14, 3.14]$ radians. This allows for a unified policy architecture while maintaining type-specific dynamics. We model the variables jointly and define their joint conditional PMF given the agent’s observation $\mathbf{o}_t^i$:

$$\pi_{\tilde{a}, \tilde{s}}(a, s, h \mid \mathbf{o}_t^i) := P(\tilde{a} = a, \tilde{s} = s, \tilde{h} = 0 \mid \mathbf{o}_t^i) \quad (3)$$

The resulting 91-dimensional action space is stepped using an Ackermann bicycle model (Rajamani, 2011).

3.4 Reward function

The reward function encourages agents to follow expert suggestions while avoiding unsafe behaviors that lead to collisions or off-road events. It consists of two components: an *internal* reward that penalizes unsafe actions, and a *guidance* reward that incentivizes agents to follow the reference trajectory and exhibit smooth dynamics.

$$R(\mathbf{o}_t^i, \mathbf{a}_t^i) = R_{\text{internal}}(\mathbf{o}_t^i, \mathbf{a}_t^i) + R_{\text{guidance}}(\mathbf{o}_t^i, \mathbf{a}_t^i, G) \quad (4)$$

where G is the oracle from which we sample guidance points. During training, we take the guidance points directly from the recorded logs. However, this framework enables flexible sampling of reference points from any distribution.

3.4.1 Internal reward

The internal reward is defined as:

$$R_{\text{internal}}(\mathbf{o}_t^i, \mathbf{a}_t^i) = w_{\text{collision}} \cdot \mathbb{I}[\text{collided}] + w_{\text{offroad}} \cdot \mathbb{I}[\text{off_road}] \quad (5)$$

Here, $\mathbb{I}[\cdot]$ is an indicator function, and $w_{\text{collision}}, w_{\text{offroad}}$ are set to -0.1, which is ten times the reward for matching an expert suggestion.

3.4.2 Guidance reward

The guidance reward is designed to encourage agents to: 1) Complete the route (x, y positions); 2) Stay close to the provided heading and speed targets; 3) Drive smoothly with minimal jerk. We create a reward term to achieve each of these objectives.

$$R_{\text{guidance}}(\mathbf{o}_t^i, \mathbf{a}_t^i, G) = R_{\text{route}} + R_{\text{dynamics}} + R_{\text{smoothness}} \quad (6)$$

Route completion Agents are rewarded for progressing along the reference positions as follows:

$$R_{\text{route}} = \text{clip} \left(\frac{\text{new_guidance_points_hit}}{\text{total_valid_points}}, 0, 0.1 \right) \quad (7)$$

We clip the reward to 0.1 since some agents already have N points within reach at reset, and we don't want to introduce inconsistencies in the rewards. Agents can only complete a single route point per tick (step). If they hit more than one point, only the closest one is marked as complete. Note that the route points are binary: an agent hits the route point if it is within a radius of 1 meter of the (x, y) point. This is done to allow some space for flexibility (akin to real-world navigation, we want agents to roughly follow the route, not necessarily *minimize* the distance to it).

Dynamics matching To encourage the agent to match the guidance speed and headings, we add a small penalty:

$$R_{\text{dynamics}} = -w_{\text{speed}} \cdot (1 - \exp(-(\hat{v} - v)^2)) - w_{\text{heading}} \cdot (1 - \exp(-(\hat{\psi} - \psi)^2)) \quad (8)$$

Where $\hat{v}$, $\hat{\psi}$ are the reference speed and heading, and v , ψ are the actual values taken by the agent. The exponential terms keep the reward bounded and provide smooth gradients. Small errors are penalized mildly, while large deviations quickly saturate the penalty. Both weights are set to 0.005.

Action smoothness To encourage smooth, more human-like control, we penalize abrupt changes in acceleration and steering:

$$R_{\text{smoothness}} = -w_{\text{smooth}} \cdot ((1 - \exp(-\Delta a^2)) + (1 - \exp(-\Delta \delta^2))) \quad (9)$$

Where Δa and $\Delta \delta$ are the differences in acceleration and steering between consecutive timesteps.

We also note that we remove the tendency to complete the route as soon as possible by only allowing one completed route point per step and setting $\gamma = 1.0$ so that rewards are not discounted (i.e., earlier is not better).

3.5 Guided self-play with stochastic dropout

We train agents using Proximal Policy Optimization (Schulman et al., 2017, PPO). To allow agents to effectively coordinate with other road users during closed-loop inference, we use a decentralized self-play set up where all agents share a single policy. Although the policy is shared, each agent operates with its own partial view of the environment and must act based on limited information about others, reflecting some of the uncertainty humans face when navigating traffic.

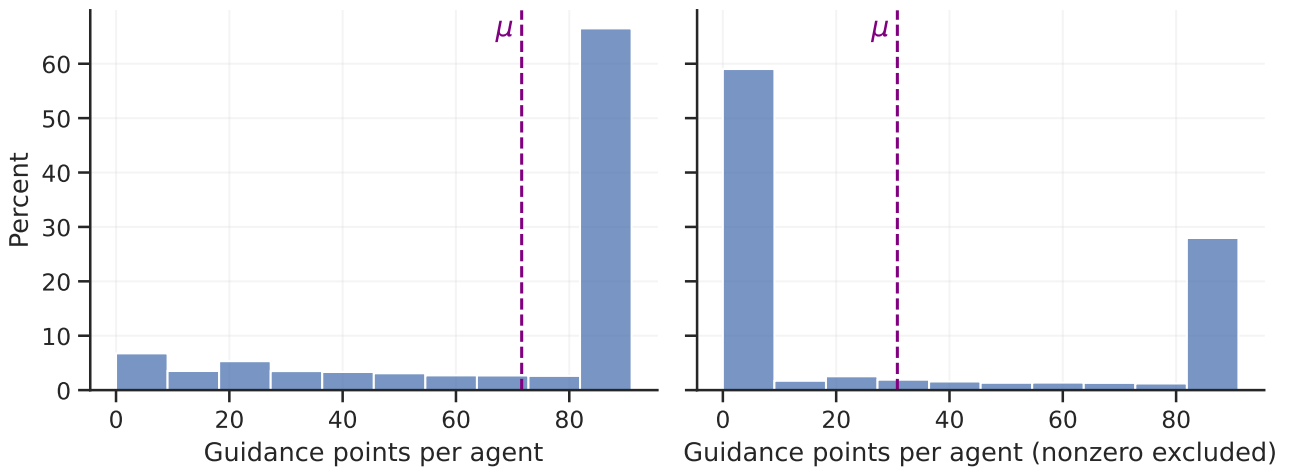


Figure 2: Natural variability in the time series of log replays in 1,000 randomly sampled scenarios from the Waymo Open Motion validation dataset. *Left:* Distribution of reference points per agent (valid time steps); *Right:* Distribution of reference points when excluding zero time series (valid and nonzero).

To reduce reliance on guidance and allow its removal at test time, we apply stochastic dropout during training. The Waymo Open Motion dataset exhibits some natural variability due to occasional invalid labels, which is an artifact of how the data was collected (Ettinger et al., 2021). However, as shown in Figure 2, most agents (over 60%) have 91 valid guidance points. The right-hand side depicts the distribution when we exclude zero-valued guidance points (typically parked vehicles), which reveals that the natural distribution of guidance points in the data is bimodal.

Since we want agents to be robust to any level of guidance, we further increase guidance variance by randomly dropping 0%–99% of guidance points during training. At each reset, a dropout fraction $\mathbf{d}_g \sim \text{Beta}(\alpha = 3, \beta = 1)$ is sampled independently for each agent, to bias the guidance towards sparse cases. The final guidance point is always retained, allowing up to 90 points to be dropped.

Agents are trained for 15 billion steps over 7 days on a single H100 GPU (across 700 parallel worlds and on 10,000 random scenarios from the WOMD train dataset). Full implementation details and hyperparameters are provided in Appendix X.

3.6 Baselines

We evaluate the effect of guidance by removing it selectively from the observation and reward function. Specifically, we compare our method against the following baselines:

- **Goal-conditioned RL (no guidance):** Self-play using only internal rewards, with no guidance input. (e.g., GigafLOW policy)
- **Stochastic dropout guidance:** Guidance is always used in the reward, but included in the observation with stochastic dropout.

4 Results

4.1 Agent realism

Benchmark: Goal-conditioned WOSAC We evaluate our method on the Waymo Open Sim Agents Challenge (WOSAC) (Montali et al., 2024), a standard benchmark for testing closed-loop agent realism in large-scale, multi-agent driving simulations. Each scenario involves up to 128 agents, including the autonomous vehicle (AV), and requires predicting $x/y/z/\theta$ trajectories over 80 steps (8 seconds). Following the WOSAC protocol, we run 32 rollouts per scenario on held-out scenes from the Waymo Open Motion Dataset (WOMD) (Ettinger et al., 2021). In a small deviation from the standard setup, which uses 1 second of logged context, we instead condition each agent on a small set of future guidance points. These points are either taken from the original logs or generated by a pre-trained Versatile Behavior Diffusion (VBD) model (Huang et al., 2024). Each agent receives its own reference trajectory in a local frame, comprising speed, heading, and position over time (see Section 3.2). Figure 7 shows the difference between the standard WOSAC protocol and our goal-conditioned setup. For comparison, we include two state-of-the-art open-sourced models: SMART (Wu et al., 2024), which autoregressively samples motion tokens with transformers, and VBD, a diffusion-based joint predictor. Both are trained on WOMD with their default configurations. We also report the performance of the logged oracle as an upper bound.

Guided self-play achieves realism on par with the state of the art with 2 % of the training data Our method achieves a WOSAC meta-score of 0.763 when conditioned only on a single last guidance point, which is similar to SMART (0.78) and it outperforms VBD (0.73). This illustrates that guided self-play with the right reward design yields agents with realistic interactive behavior competitive with current best-performing models. Table 1 shows the composite and category metrics. The full breakdown of metrics per category are shown in Tables 5, 6, 7.

Comparable performance with smaller models and faster runtime As shown in Figure 3, our model achieves similar WOSAC meta-scores while requiring significantly less computation. With only 1.3 million parameters, it

is 6–10 $\times$ smaller than the baselines and runs faster per scenario. This suggests that guided simulation agents without sacrificing performance.

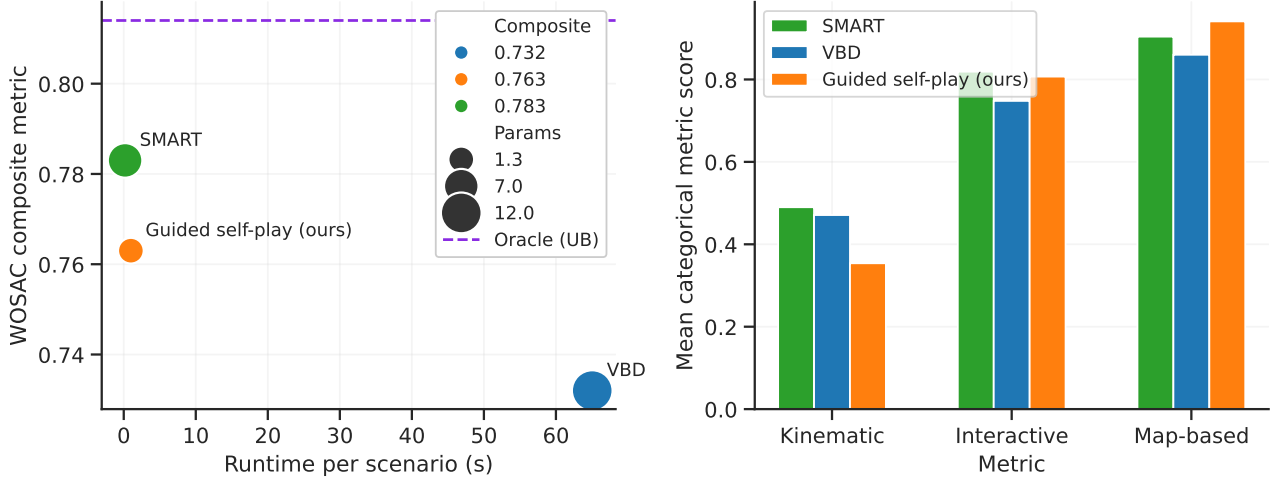


Figure 3: [PLACEHOLDER:Model still running] Benchmarking our guided self-play method to two state-of-the-art supervised approaches. *Left*: WOSAC composite metric as a function of the runtime per individual scenario. The size of the dot is proportional to the number of model parameters. *Right*: Breakdown of WOSAC scores between the main categories.

Table 1: Aggregate meta-score and per-category WOSAC metric results on 229 scenarios from the WOMB validation-interaction split.

Method	Model Size	Composite $\uparrow$	Kinematic $\uparrow$	Interactive $\uparrow$	Map-Based $\uparrow$	minADE[m] $\downarrow$	Collision Rate	Offroad Rate
SMART	7 M	0.7838	0.490	0.819	0.904	1.17	0.008	0.027
VBD	12 M	0.732	0.471	0.748	0.860	1.734	0.107	0.108
Guided self-play (ours)	1.3 M	0.763	0.354	0.807	0.941	2.708	0.003	0.05
Logged oracle	N/A	0.814	0.558	0.842	0.924	N/A	0	0.022

Realism is maintained across full range of guidance densities Figure 4 shows that the WOSAC realism meta-score remains stable across different guidance densities. The score decreases by only 0.01 when using a single guidance point, indicating minimal performance degradation. Figure 5 depicts the same analysis, but then broken down by individual WOSAC category metrics, confirming that realism is preserved even under sparse conditioning.

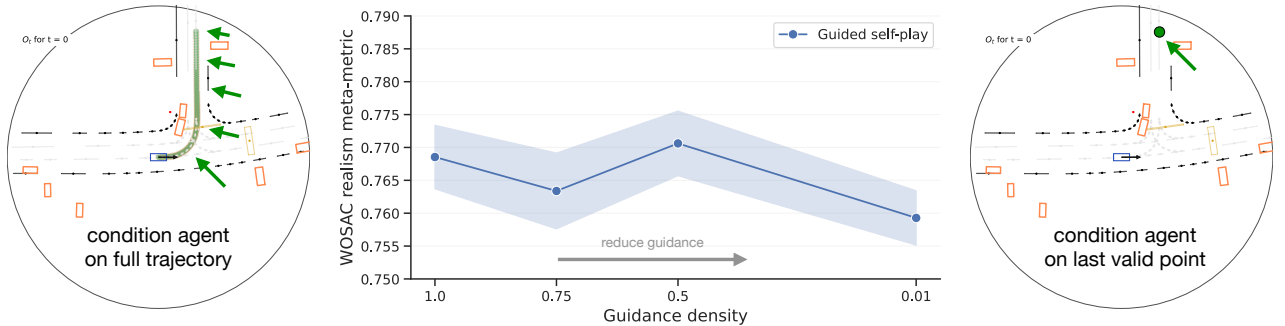


Figure 4: WOSAC realism meta-metric as a function of the guidance density. We plot the mean and standard error of the WOSAC realism meta-score as a function of the guidance density, that is, the ratio of guidance points per episode. At guidance ratio = 1.0 (left), an agent receives a guidance point (x, y, s, ψ) for every time step during the scene rollout (at most 90 steps). At guidance density 0.01, we remove all guidance points except the very last valid point. The realism meta-score ranges between 0.77 and 0.76. We use 229 interactive validation scenes for evaluation.

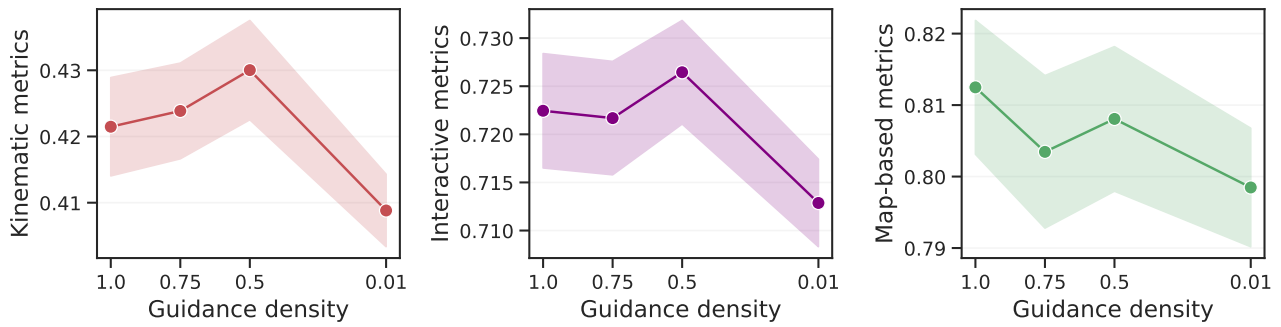


Figure 5: WOSAC category metrics as a function of guidance density. Mean and standard error for 229 held-out scenarios from the WOMB interactive validation dataset.

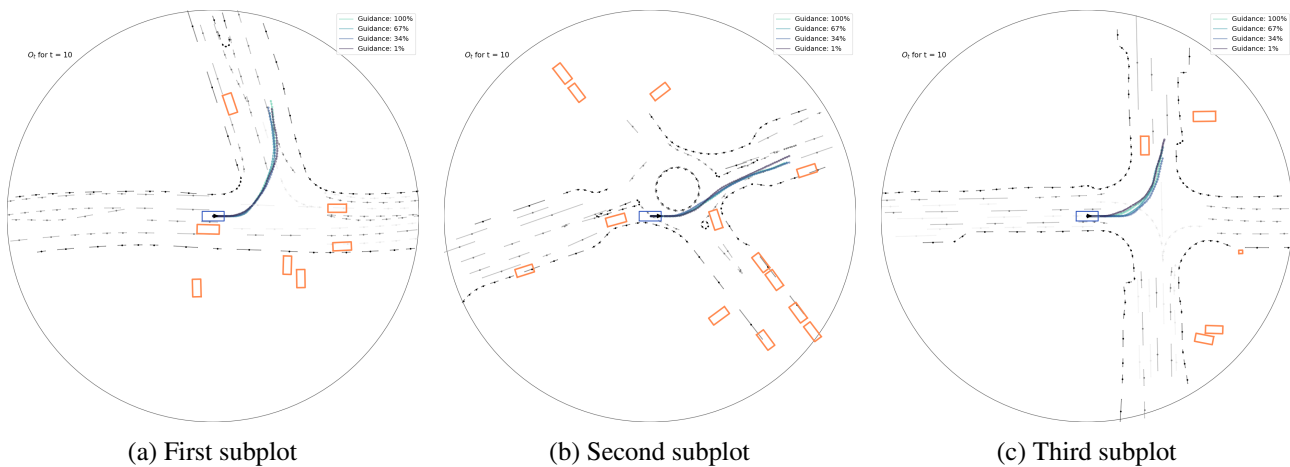


Figure 6: Example rollouts with different levels of guidance: trajectory is consistent.

Q2: What is the agent realism (performance) when we remove any guidance at all?

We also explore what happens when we remove all guidance points from the agent’s observation. Removing all guidance points essentially removes the goal of the agents.

4.2 In-depth analysis

Error distributions

Example behaviors and rollout plots

4.3 Agent realism when conditioning on unseen guidance points

Q5: How does the agent realism (performance) change when we take the guidance points from a model instead of directly from the data (out of distribution)

What if we don’t have access to the logs?

Q6: Qualitative analysis of behavior difference as a function of guidance density

Q7: Is there a difference in performance between agent types?

a) Table with aggregate wosac scores, summarize performance; b) Compare and contrast with reliable agents/GIGAFLow policy

Distribution of scores across scenarios It may be interesting to plot the distribution of meta-scores across $N=100$ or 1000 scenarios, and learn about what kind of scenarios yield good/bad scores. It might also be

informative to know how the distribution looks like (e.g. are we doing equally good/bad on all scenes, or are there a few scenes that are very difficult?)

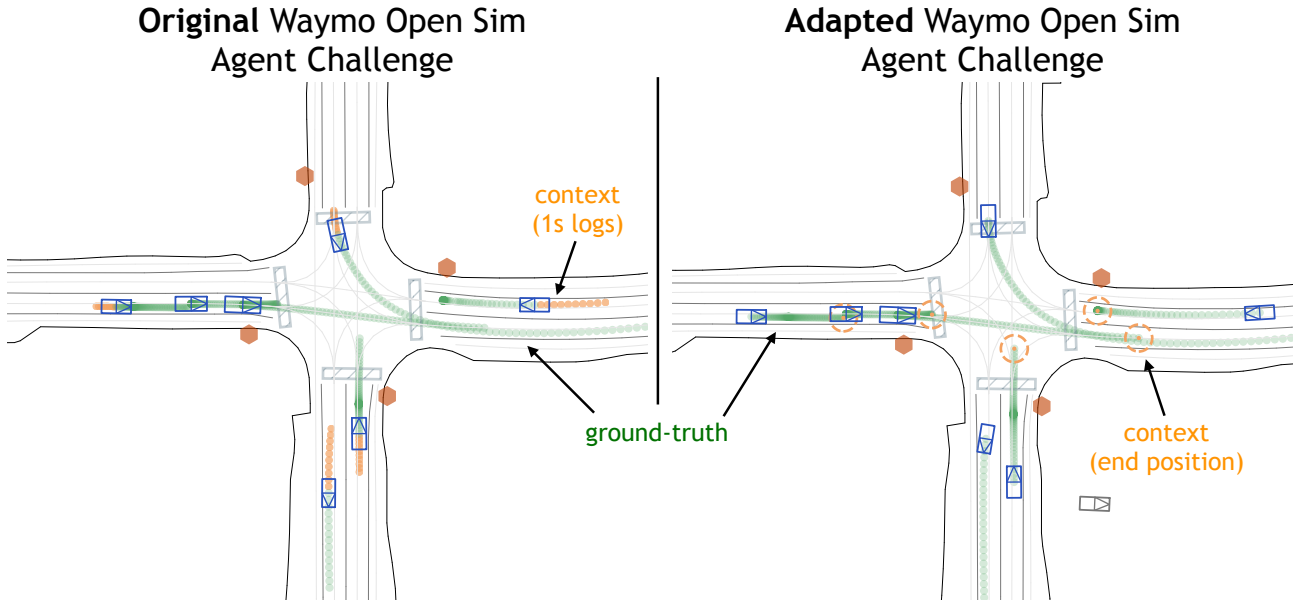


Figure 7: Waymo Open Sim Agent Challenge. We evaluate our approach in two versions of the Waymo Open Sim Agent Challenge (WOSAC). *Left*: The goal of WOSAC is, given the first second of context in a scenario, to predict the joint behaviors of all, up to 128 agents in the scenario. A high score is achieved when the distribution (32 rollouts) of predicted trajectories is close to the actual ground-truth for each particular agent. *Right*: In addition to the standard WOSAC benchmark, we also evaluate agents in an adapted version where we assume that we have access to a high-level goal position of the agents, and instead focus on the distribution of the navigation to these high-level targets. Instead of receiving some context (past behavior), agents do not receive this but are instead conditioned on the high-level joint destinations of the agents. Here, we essentially assess whether agents behave in a human-like way without the additional context. This is useful because if successful, it means that we can place high-level targets in the scene, and agents will navigate there in a realistic way.

5 Discussion

DC: Write, below are just some notes

Data scaling laws for guided self-play not explored here, needs to be systematically studied.

Conditioning on the future versus the past Rather than conditioning agents on 1 second of past context—as is standard in WOSAC—we condition on a small number (up to one) of future guidance points. This assumes that agents are operating with a known high-level intention, such as a target destination. This shift reframes the problem from distribution matching to goal-directed human-like control.

Learning in decentralized settings Unlike most competitive WOSAC submissions, which operate with centralized policies and access to the full scene state, our method learns a decentralized policy using only local observations and a guidance signal. While centralized policies can exploit global information to improve prediction accuracy, they are less representative of human road behavior in the sense that agents must act with limited and noisy inputs. Our results suggest that high levels of realism are achievable even under more realistic information constraints.

6 Conclusion

Contribution statement

Write up contributions

DC

KJ

ZZ

JFF

EV

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A Dataset

B Observation features and design choices

The observation at time step t for agent i , $\mathbf{o}_t^i$, is multimodal and consists of four types of information: the ego state, expert guidance, the road graph, and the partner observation. We set the maximum number of agents per scenario throughout the experiments, $N = 32$. A given agent’s observation is provided as a flattened vector of 2974 elements.

Table 2: Ego state features and dimensions provided in the observation $\mathbf{o}_t^i$.

Feature	Dimension	Description
Speed	1	The speed of the agent
Vehicle length	1	Length of the agents’ bounding box
Vehicle width	1	Width of the agents’ bounding box
Steering angle	1	The current steering wheel angle of the agent
Steering angle	1	The current steering wheel angle of the agent

Table 3: Visible view or road graph features and dimensions provided in the observation $\mathbf{o}_t^i$. The road graph consists of a sampled set of R nearest road points, where R is set to 200 in the experiments.

Feature	Dimension	Description
x	$1 \cdot R$	Relative x coordinate of the road point
y	$1 \cdot R$	Relative y coordinate of the road point
Segment length	$1 \cdot R$	Length of the road segment associated with the (x, y) coordinate
Segment width	$1 \cdot R$	Width of the road segment associated with the (x, y) coordinate
Segment height	$1 \cdot R$	Height of the road segment associated with the (x, y) coordinate
Segment orientation	$1 \cdot R$	Angle between the segment midpoint and the ego agent
Type	$1 \cdot R$	Integer indicating the type of the road point. Existing types are: Road edge (impassable; boundary of the road), road lane, road line, stop sign, crosswalk, and speed bump. Integers are one-hot encoded during training, which multiplies the feature dimension by the total number of classes.

Table 4: Partner (“the other”) agent features and dimensions provided in the observation $\mathbf{o}_t^i$. Partner information is visible within the observation radius.

Feature	Dimension	Description
Speed	$1 \cdot N - 1$	The speed of the other agents
$(\mathbf{x}, \mathbf{y})$	$2 \cdot N - 1$	Relative positions of the other $N - 1$ agents in the scene. Information is only provided if the partner agents are within the observation radius of the ego agent, and are left as zero otherwise.
(θ_x, θ_y)	$2 \cdot N - 1$	Relative orientation of the other $N - 1$ agents in the scene. Information is only provided if the partner agents are within the observation radius of the ego agent, and are left as zero otherwise.
(w, l, h)	$3 \cdot N - 1$	The width, length, and height of the bounding boxes of the other agents.

Guidance signals We also smooth the velocities and headings

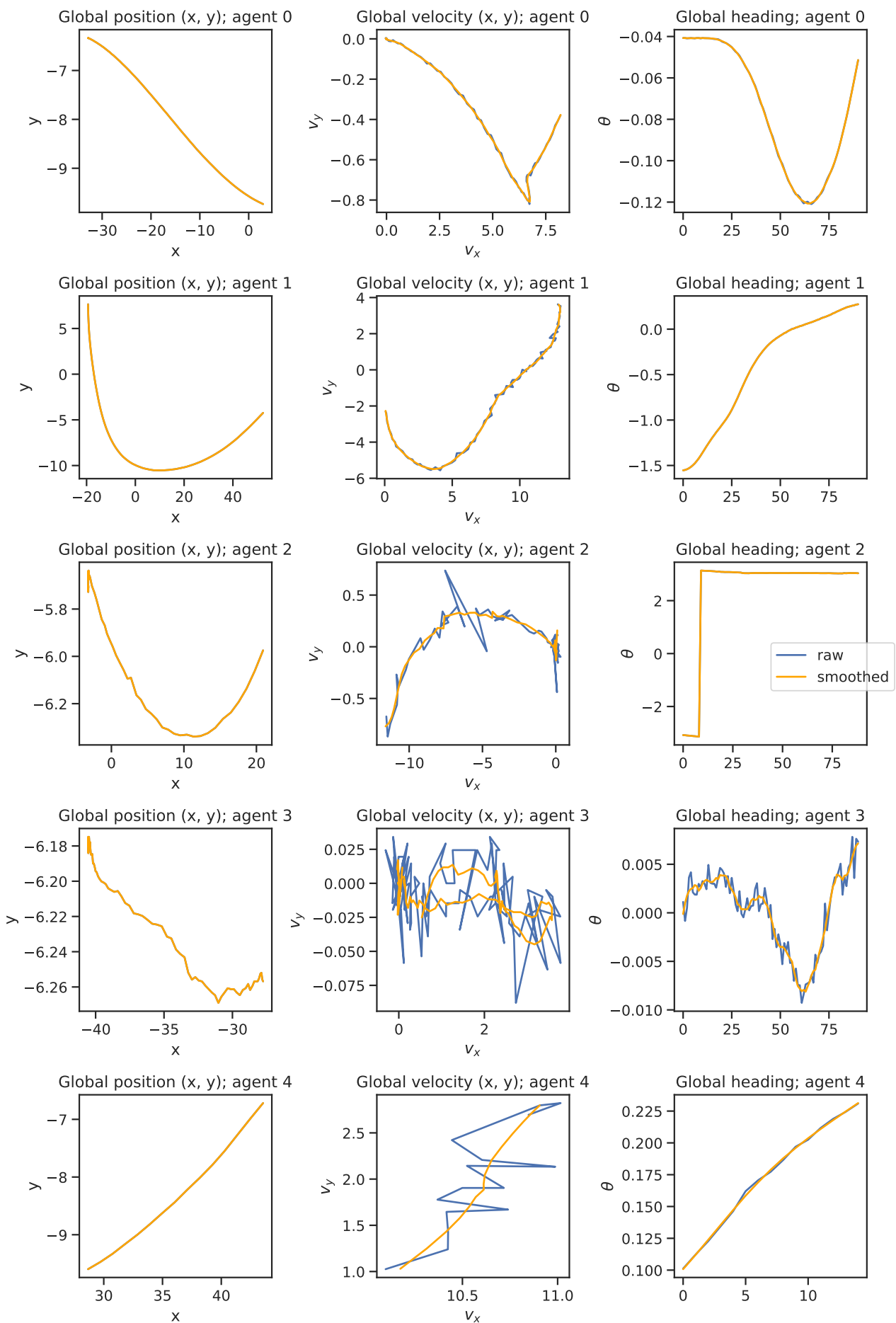


Figure 8: Example trajectories.

C Waymo Open Sim Agent Challenge Additional Tables

Table 5: Kinematic WOSAC metric breakdown scores on 229 scenarios from the WOMB validation-interaction split.

Method	Kinematic Score $\uparrow$	Linear Speed $\uparrow$	Linear Accel $\uparrow$	Angular Speed $\uparrow$	Angular Accel $\uparrow$
SMART	0.4919	0.3538 $\pm$ 0.1659	0.4639 $\pm$ 0.1123	0.4182 $\pm$ 0.2306	0.7317 $\pm$ 0.1478
VBD	0.4712	0.3248 $\pm$ 0.176	0.4612 $\pm$ 0.1252	0.3747 $\pm$ 0.2434	0.7241 $\pm$ 0.1751
Guided self-play (ours)					
Logged Oracle	0.5584	0.4699	0.5258	0.469	0.7688

Table 6: Interaction WOSAC metric breakdown scores on 229 scenarios from the WOMB validation-interaction split.

Method	Interactive Score $\uparrow$	Dis. to Nearest Object $\uparrow$	Collision Indication $\uparrow$	Time to Collision $\uparrow$	Collision Rate $\uparrow$
SMART	0.8206	0.3111 $\pm$ 0.1907	0.9902 $\pm$ 0.0646	0.9062 $\pm$ 0.1626	0.0091 $\pm$ 0.0608
VBD	0.748	0.2808 $\pm$ 0.1961	0.8773 $\pm$ 0.2352	0.8917 $\pm$ 0.1793	0.107 $\pm$ 0.2049
Guided self-play (ours)					
Oracle	0.8417	0.3655	0.999	0.9246	0

Table 7: Breakdown of WOSAC Map-based metric results on 229 interactive scenarios from the WOMB validation-interaction split.

Method	Map-based Score	Dis to Road Edge	Offroad Indication	Traffic Light Violation	Offroad Rate
SMART	0.9042	0.6926 $\pm$ 0.2149	0.9889 $\pm$ 0.0939	0.9999 $\pm$ 0	0.024 $\pm$ 0.1084
VBD	0.8599	0.6529 $\pm$ 0.2334	0.8732 $\pm$ 0.2601	0.9999 $\pm$ 0	0.1085 $\pm$ 0.1977
Guided self-play (ours)					
Oracle	0.9238	0.7359	0.999	0.999	0.0218

D Reflections on the Waymo Open Sim Agent Challenge: Assumptions and limitations

The Waymo Open Sim Agents Challenge (WOSAC) serves as a benchmark for behavior prediction, but its design does not necessarily align with the goal of producing autonomous, human-like behavior. The core task is framed as a distribution-matching problem: given a set of 32 model-generated rollouts, WOSAC computes the per-agent likelihood of human trajectories under the model’s predicted distribution. The meta-score reflects the question: *How probable would these human samples be under the learned model?*.

This formulation favors low-entropy models that concentrate probability mass around the observed human behavior. As a result, models can achieve high scores by closely reproducing the exact logged trajectories. For instance, earlier goal-conditioned self-play reinforcement learning (RL) agents (Cusumano-Towner et al., 2025) perform well by simply reaching the end (x, y) positions while avoiding collisions. In some cases, matching parked vehicles was achieved by setting their velocities to zero, effectively reproducing their stationary trajectories.

D.1 Key Assumptions of WOSAC

Dependence on Human Data Context. A key assumption is the availability of sufficient human context data. WOSAC models rely on the 1-second input context. This is reasonable when simulating counterfactuals based on real-world scenarios, and in most real-world driving there is always some information available to condition on.

Single Ground-Truth Assumption. WOSAC implicitly assumes a single ground-truth behavior per agent in a given context. This ignores the inherently multi-modal nature of human navigation, where behavior can vary due

to factors like weather, time of day, or individual preferences. While one could argue that this issue diminishes with enough data, we currently lack a quantitative understanding of behavioral diversity in datasets like WOMD.

Short-Term vs. Long-Term Consistency. The benchmark assumes that optimizing likelihood over short (9-second) trajectory snippets will lead to globally consistent behavior. It remains unclear whether this assumption holds in practice, particularly when modeling long-term interactions or planning.

Rare Event Handling. Finally, rare or unusual events are likely underrepresented in the dataset. Yet these scenarios are critical for evaluating safety and robustness in autonomous systems. A model optimized solely for WOSAC performance may fail to generalize to such edge cases.

E Technical Appendices and Supplementary Material

Technical appendices with additional results, figures, graphs and proofs may be submitted with the paper submission before the full submission deadline (see above) or as a separate PDF in the ZIP file below before the supplementary material deadline. There is no page limit for the technical appendices.